

PETERBOROUGH, ON, Canada

WMO: 714360

Lat: 44.2303N

Lon: 78.3526W

Elev: 191

StdP: 99.05

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-20.3	-28.9	0.3	-23.3	-25.3	0.4	-19.4	11.5	-3.0	9.9	-3.9	1.7	310	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	30.1	22.3	28.5	21.3	27.1	20.5	23.7	28.1	22.6	26.7	21.7	25.4	4.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.3	26.1	21.3	16.3	24.8	20.4	15.4	23.9	71.8	28.3	67.7	26.7	64.2	25.4	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.1	7.1	-28.4	32.6	3.9	2.2	-31.2	34.2	-33.5	35.5	-35.6	36.8	-38.4	38.4		
(5)			-28.6	25.6	3.8	1.1	-31.4	26.4	-33.6	27.0	-35.8	27.7	-38.5	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-7.5	-6.4	-1.4	5.9	12.7	17.5	19.9	18.9	14.9	8.4	2.1	-3.6	(6)
(7)		DBStd	10.77	7.02	6.21	6.02	4.49	4.22	3.50	2.92	2.94	4.01	4.37	4.74	5.80	(7)
(8)		HDD10.0	2261	543	458	355	137	20	1	0	0	6	84	238	420	(8)
(9)		HDD18.3	4347	801	692	612	373	181	56	17	28	117	307	486	678	(9)
(10)		CDD10.0	1119	0	0	2	14	104	227	307	277	153	36	2	0	(10)
(11)		CDD18.3	164	0	0	0	0	7	32	65	47	13	1	0	0	(11)
(12)		CDH23.3	1718	0	0	1	6	102	336	661	445	159	8	0	0	(12)
(13)		CDH26.7	394	0	0	0	0	18	78	170	96	32	0	0	0	(13)
(14)	Wind	WSAvg	2.9	3.4	3.4	3.5	3.5	3.0	2.6	2.4	2.2	2.2	2.8	3.0	3.3	(14)
(15)	Precipitation	PrecAvg	888	72	52	56	75	85	82	84	69	78	73	82	75	(15)
(16)		PrecMax	1184	133	98	101	146	133	195	249	136	173	167	152	132	(16)
(17)		PrecMin	698	41	21	19	27	23	29	33	28	38	26	47	18	(17)
(18)		PrecStd	123	23	22	25	31	28	46	46	28	30	35	28	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	9.7	19.8	24.0	28.9	30.7	32.2	31.9	30.1	24.5	17.1	12.0	(19)
(20)			MCWB	8.9	7.3	14.0	16.4	20.9	21.8	24.0	22.6	22.7	19.2	12.4	10.6	(20)
(21)		2%	DB	6.2	6.3	13.4	20.2	26.1	28.9	30.3	29.1	27.2	21.2	14.0	8.4	(21)
(22)			MCWB	4.9	4.5	9.4	12.8	18.8	21.5	22.8	21.7	20.9	16.4	11.7	7.1	(22)
(23)		5%	DB	3.8	3.9	9.8	17.0	23.7	27.1	28.5	27.3	25.1	18.7	12.0	6.0	(23)
(24)			MCWB	2.7	2.4	6.4	10.9	16.8	20.5	21.4	20.7	19.5	15.1	9.5	4.9	(24)
(25)		10%	DB	1.9	2.2	7.0	14.2	21.3	25.1	27.0	25.9	22.9	16.2	9.8	3.9	(25)
(26)			MCWB	0.9	0.9	4.2	9.0	15.3	19.1	20.5	19.9	18.3	13.4	7.8	2.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	7.5	14.7	17.1	21.8	24.0	25.5	24.6	23.5	20.2	13.7	10.8	(27)
(28)			MCD B	9.8	8.8	18.5	22.2	27.3	29.1	30.5	28.7	27.9	22.8	15.2	11.6	(28)
(29)		2%	WB	5.2	4.8	9.8	14.3	19.9	22.6	24.0	23.1	22.1	17.3	12.0	7.4	(29)
(30)			MCD B	5.9	6.0	12.3	18.5	25.0	27.5	28.5	27.0	25.7	20.2	13.7	8.2	(30)
(31)		5%	WB	2.9	2.6	6.9	11.8	18.1	21.4	22.7	22.2	20.6	15.5	9.8	5.0	(31)
(32)			MCD B	3.8	3.7	9.5	15.8	22.2	25.5	26.7	25.7	23.7	18.0	11.6	6.0	(32)
(33)		10%	WB	0.9	1.1	4.4	9.7	16.3	20.2	21.6	21.1	19.2	13.7	7.8	2.8	(33)
(34)			MCD B	1.7	2.0	6.7	13.4	20.0	23.5	25.3	24.3	22.0	15.7	9.7	3.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.5	10.0	10.0	11.0	12.4	11.9	12.5	12.5	13.0	10.6	8.8	7.8	(35)
(36)			MCD BR	8.6	10.5	14.0	16.7	15.8	14.5	14.1	14.5	15.5	14.5	12.7	9.2	(36)
(37)			MCWBR	8.1	9.1	10.0	10.4	8.7	8.0	7.4	7.9	9.3	10.1	10.2	8.3	(37)
(38)		5% WB	MCD BR	8.7	9.8	13.0	14.1	13.8	12.4	12.3	12.0	13.2	13.1	11.6	8.8	(38)
(39)			MCWBR	8.4	8.9	10.1	10.1	8.6	7.5	7.2	7.3	8.6	9.9	9.9	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.305	0.319	0.355	0.378	0.413	0.413	0.403	0.365	0.351	0.309	0.283	(40)	
(41)		taud	2.326	2.329	2.419	2.381	2.361	2.284	2.295	2.315	2.414	2.442	2.509	2.455	(41)	
(42)		Ebn at Noon	811	883	920	908	892	858	854	850	859	822	811	811	(42)	
(43)		Edh at Noon	80	96	101	114	120	130	127	121	101	86	68	65	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.34	2.22	3.53	4.45	5.40	5.80	5.92	5.17	4.13	2.42	1.50	1.07	(44)	
(45)		RadStd	0.14	0.20	0.33	0.47	0.57	0.47	0.44	0.26	0.36	0.22	0.19	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page